

How Fuel Subsidy Reductions Reshaped Fishing Activity: Evidence from China's Policy Reform in the East China Sea

Jorge Montero-Mestre *

Jorge Marco.

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Abstract

This study examines the impact of fuel subsidy provision on fishing activity and unauthorized fishing following a 2016 reform to China's fuel subsidy program for its fishing fleet, focusing on how fishers responded to the differential policy change. The reform reduced the total average subsidy payment by 69%, with larger vessels receiving 15.8% more relative to smaller vessels. This reform was driven by concerns about excessive government spending, high fuel consumption, and overfishing. We develop a theoretical model to explain these dynamics and validate its predictions using a Difference-in-Differences approach. Our findings show that total fishing decreased, while unauthorized fishing hours increased in 16.6% after the reform, particularly in Taiwan and in disputed maritime zones. Furthermore, vessels that received lower subsidies were 2.5% more likely to exit the fishing sector, while those that remained increased their fishing-to-subsidy ratio as a strategy to compensate for the loss of competitiveness.

Key words: Commercial Fishing, Fisheries Management, Fuel Subsidies, Unauthorized Fishing.

JEL Classification: H23,O13,Q22,Q28.

* *Corresponding author:* **Jorge Montero** is a Postdoctoral Fellow. Department of Economics, Universidad de los Andes, Cra 1, No. 18A-12, postal code 111711, Bogotá, Colombia (email: jl.montero@uniandes.edu.co). **Jorge Marco** is an Assistant Professor. Department of Economics, University of Girona, Spain (email: jorge.marco.renau@udg.edu). The authors would like to thank Jorge H. Maldonado, Sandra Aguilar, Andries Richter, BlueRforD members and ELG 2025 conference participants for their valuable comments.

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1 Introduction

Input subsidies have been widely used as an instrument for capacity-enhancement across various sectors of the economy (Reyes-García et al., 2025). In the fisheries sector, in particular, there has been considerable debate about the implications of subsidies in contributing to the overexploitation of marine resources and the depletion of natural products (Sumaila et al., 2021, Shen and Chen, 2022, Finley, 2017), especially in common-pool settings (Munro and Sumaila, 2002). In 2018, approximately 35.4 billion dollars were distributed in subsidies to global fisheries, of which capacity-enhancing subsidies are USD 22.2 billion (Sumaila et al., 2019). This has generated overcapacity leading to overexploitation problems (Sumaila et al., 2010a), especially in scenarios with low controls (Wang et al., 2023, Skerritt et al., 2023).

According to FAO (2024), the fraction of fishery stocks outside biologically sustainable levels increased to 37.7 percent in 2021, that is 2.3 percent higher than in 2019, raising alarm over the accelerating degradation of marine ecosystems and the sustainability of global fisheries. Thus, the World Trade Organization (WTO) has been promoting negotiations to eliminate capacity-enhancing subsidies in fisheries to ensure the reduction of overfishing levels (Cisneros-Montemayor et al., 2022), to achieve the Sustainable Development Goal (SDG) 14.6 (Shen and Chen, 2022, Skerritt and Sumaila, 2021, Yang et al., 2017). The provision of subsidies is particularly concerning because it allows vessels to fish outside their own countries, venturing into the high seas or waters of other countries (Englander et al., 2025), generating rents that would not have been possible without subsidies (Sumaila et al., 2019, Sakai et al., 2019, Sumaila and Pauly, 2013). Moreover, this creates distortions in national budgets, complicating the management of other fiscal objectives and increasing the opportunity cost for governments and society (Davis, 2014).

Despite recent causal evidence on how subsidy provision influences fishing capacity (Wang et al., 2023, Shen and Chen, 2022) and overfishing (Englander et al., 2025), there remains limited empirical causal analysis on how access to subsidies shapes incentives for unauthorized fishing activity and on the behavioral adjustments of vessels facing relative subsidy reductions. This study examines how differential changes in fuel subsidy provision and exit incentives from a Vessel Buyback Program (VBP) influence fishing effort and the incidence of Illegal, Unreported, and Unregulated (IUU) fishing by Zhejiang Province’s industrial fishing fleet, particularly in ecologically and geopolitically sensitive waters such as those of Taiwan, Japan, and Disputed Zones¹? IUU fishing is defined here as unauthorized fishing by vessels operating in foreign waters without formal access agreements, a practice considered an environmental injustice for its externalization of costs onto other nations’ marine resources. This subsidy provision has been shown to contribute to other social problems such as piracy (Phayal et al.,

¹Disputed zones are usually maritime areas claimed by two or more countries where their Exclusive Economic Zones (EEZs) overlap. They often lead to tensions as each nation asserts its own sovereign rights and, therefore, the right to exploit the disputed territory (Østhagen, 2020).

2024, Mitchell and Schmidt, 2024, Denton and Harris, 2021).

This paper examines the 2016 Chinese subsidy reform as a case study, which implemented a differential reduction in fuel subsidies across vessels based on their lengths, coupled with the existence of the VBP, representing the opportunity cost of remaining in the fishing sector. We focus on China case because is the largest provider of fisheries subsidies worldwide. In 2018, it allocated USD 7.2 billion, representing 21% of global fisheries subsidies, 81% of which were classified as capacity-enhancing (Sumaila et al., 2019). This level of support has contributed to the expansion of China’s fishing capacity, raising international concern over its impact on the sustainability of marine ecosystems (Englander et al., 2025). In addition, China’s extensive involvement in IUU fishing has become a source of diplomatic tension (Tseng and Ou, 2010, Zhang, 2016, Kim, 2019, Watson and Woodill, 2022). Zhejiang province plays a central role in this context. It hosts the largest fishing fleet in the East China Sea, with a significant share of vessels operating through trawling methods (Ministry of Agriculture Fisheries Administration Bureau, 2006), one of the most destructive practices for marine ecosystems (Buhl-Mortensen and Buhl-Mortensen, 2018). Furthermore, it is governed by China’s most prominent coastal authority (Wang et al., 2023), making it a strategically important area for examining the effects of subsidy reductions on fishing activities, including IUU fishing in Disputed Zones and the surrounding regions.

Our analysis focus on changes in subsidy provision, which can led to either a direct response in fishing activity through cost structure adjustments or an inverse response where vessels attempt to offset losses in competitiveness by modifying behavior (Sakai et al., 2019). This is especially relevant when subsidy reductions create incentives affecting decisions related to IUU fishing. The concerns surrounding IUU fishing extend beyond fish stock depletion (Khan et al., 2024) to include impacts on international relations among countries (Spijkers et al., 2019, Chen et al., 2023).

In this paper, we develop a theoretical model to analyze how subsidy reforms can affect fishing activity by modifying cost structures and shaping fishers’ expectations. The predictions of the model related to unauthorized fishing are evaluated using a Difference in differences (DiD) approach that draws on a newly released dataset compiled by Wang et al. (2023), which includes individual-level data on trawler fishing vessels from the Marine Fishing Vessel Dynamic Management System of Zhejiang province. To capture real-time fishing activity, we incorporate data from Global Fishing Watch (GFW). To classify fishing operations by authorization status, we rely on information from the Fishing Access Agreements database maintained by the Sea Around Us platform.

Our results suggest that, following the reform, total fishing effort declined significantly, with the largest reduction among smaller vessels. This pattern reflects both their response to lower subsidy provision (15.8%) and the decline in fishing capacity driven by the VBP (2.5%). Moreover, vessels receiving larger subsidies reported higher levels of unauthorized

fishing (16.6%), particularly in Taiwanese waters and Disputed Zones. The increase in IUU fishing among larger vessels can be explained by their greater physical capacity to undertake longer fishing trips, as well as by their ability to capture the IUU opportunities previously exploited by smaller vessels before the reform. The reform reduced the fishing capacity of smaller vessels, which had historically engaged more intensively in IUU fishing due to the overcapacity created by subsidies.

We also find that compliance levels² in China are high compared to the global average (95.1% vs. 61%; see [Montero-Mestre and Maldonado \(2025\)](#)). The reform’s design—excluding noncompliant fishers from receiving subsidies—combined with the VBP, likely explains these relatively high compliance levels. By contrast, the effect of subsidies on IUU fishing may be more pronounced in regions where enforcement mechanisms are weaker.

In terms of the subsidy-to-effort ratio, we observe that for every RMB/kW of subsidy received, smaller vessels exerted greater fishing effort—both total and unauthorized—than before the reform, and even more than larger vessels. This suggests that, in terms of policy effectiveness, vessels did not adjust their behavior proportionally to the reduction in subsidies. Worse still, in some cases, the response involved increasing unauthorized fishing efforts as a way to compensate for the loss of competitiveness.

This paper contributes to the literature on the effects of input subsidies—specifically the impacts of differential reductions in subsidies payments—by quantifying the relationship between fuel subsidy reform and fishing activity, with particular attention to IUU fishing ([Englander et al., 2025](#), [Wang et al., 2023](#), [Sumaila et al., 2021](#), [2010a](#)). Our results highlight the complex interplay between subsidy allocation and fishing activity, revealing that changes in the cost structure could incentivize illegal practices. The significant increase in fishing efforts in foreign waters may indicate a strategic shift among subsidized vessels seeking to maximize their competitiveness and profit margins. The correlation between subsidy levels and unauthorized fishing underscores the need for enhanced regulatory frameworks and monitoring mechanisms to ensure that financial incentives do not undermine sustainable fishing practices.

The remainder of the paper is structured as follows. Section 2 provides a description of the case study, including background information and the institutional context of the fuel subsidy reform. Section 3 introduces the theoretical model. Section 4 describes the data and the empirical strategy. Section 5 presents the results on the effects of the reform. Section 6 discusses the findings and concludes.

²We define compliance as the proportion of fishing activity conducted by vessels within the jurisdiction of other countries with proper authorization, following [FAO \(2023\)](#).

2 Study Case

2.1 Background

The East China Sea has long been a region of historical disputes between neighboring countries over sovereignty and the rights to exploit marine resources (Midford and Østhagen, 2024, Valencia, 2014). Chinese fishers have increasingly become central figures in maritime conflicts within these contentious regional waters. China’s governmental policies and strategies have directly or indirectly influenced the presence of Chinese vessels in Disputed Zones, thereby contributing to the escalation of tensions between the countries involved (Zhang, 2016). In many instances, China’s maritime territorial claims diverge from those recognized under international law (Radio Free Asia, 2024), which partly explains the persistent presence of Chinese vessels in these contested areas.

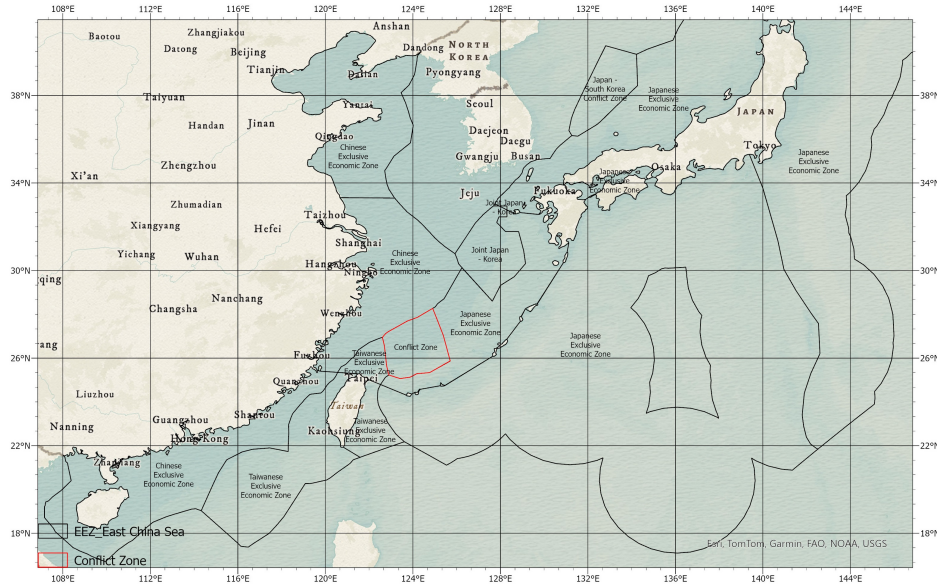


Figure 1. The East China Sea Map. Source: Authors, using information from [Marine Regions Repository](#). Note: The black solid line represents the EEZ boundaries of each country in the East China Sea, while the red solid line marks the Disputed Zone between Japan and China.

In recent years, the expansion of Chinese fishing fleets has emerged as one of the primary threats to overexploitation and unauthorized fishing in the waters of the Sea of Japan. Between 2018 and 2020, Chinese vessels were detected, on average, approximately 4,341 times within Japanese territorial waters, with a significant concentration of incidents occurring in the Disputed Zones around the Diaoyu/Senkaku Islands activities in contested maritime regions (Sato and Chadha, 2022, Reuters, 2024). One of the main explanations for this trend is that these vessels are often en route to North Korean waters, where fishing agreements and authorizations are in place. This situation underscores the complex geopolitical and environ-

mental challenges posed by the increasing presence of Chinese fishing. Another focal point of tension has been the presence of fishing activities in Taiwanese territory, particularly in the territorial waters around Kinmen, which extend approximately halfway to the Chinese coast, or roughly 4 km (2.5 miles) to the north and northwest ([Radio Free Asia, 2024](#)). Figure 1 shows the territorial waters of each country.

Regarding the fishing access relationships between China, Taiwan, and Japan, there are currently no formal agreements in place between them. This absence of agreements can be attributed to the concerns expressed by the Japanese and Taiwanese governments regarding the sustainability of their territorial waters. Moreover, the historical context of diplomatic relations among these countries, combined with China’s fisheries policies, has further complicated efforts to establish formal access agreements ([Chen et al., 2024](#)).

The lack of formal fishing access agreements is significant, particularly considering the ongoing territorial disputes in the East China Sea, where Taiwan and Japan have established a bilateral fisheries agreement. This agreement, signed on April 10, 2013, allows for reciprocal fishing rights in overlapping exclusive economic zones, thereby isolating China in the context of these maritime disputes. The agreement reflects a strategic approach by Taiwan and Japan to manage their fishing interests while setting aside sovereignty issues to foster cooperation and reduce tensions in the region ([The Diplomat, 2018](#)).

The presence of Chinese vessels in these territories is largely explained by their significant overcapacity ([Wang et al., 2023](#)) and by policies aimed at strengthening China’s food security ([Zhang, 2016](#)). These factors have incentivized the expansion and growing prevalence of Chinese fishing activities in many foreign waters. Combined with the provision of subsidies, this has stimulated greater fishing effort and heightened tensions between countries, particularly when such activities occur outside the framework of international agreements. This context of maritime tensions, lack of formal access agreements, and continued subsidy provision makes the East China Sea a particularly relevant case study for examining how changes in subsidies shape fishers’ decisions to engage in unauthorized fishing.

2.2 Institutional and Policy Overview

Due to the scale of its fishing operations, China significantly influences marine ecosystems and global fishery dynamics. Its extensive fishing history has positioned China as the nation with the highest fishing capacity in the world ([Shen and Heino, 2014](#)). China’s fishing capacity has steadily grown since 1949 and began to be regulated in 1979 in response to growing concerns over the overexploitation of marine resources ([Su et al., 2020](#)). Today, most fish stocks in China’s seas are either fully exploited or already depleted ([Teh et al., 2017](#), [Yu and Yu, 2008](#)).

Despite policy efforts to curb fishing capacity, it continues to increase. Between 2012 and 2014 alone, a 149% rise in capacity was recorded ([Yang et al., 2017](#)). One of the main challenges faced by the Chinese government is underreporting, primarily driven by the fact

that nearly all taxes, fees, and fines imposed on fishing vessels are calculated based on engine horsepower. As a result, fishers tend to underreport their vessels' horsepower, contributing to the growth of a "hidden" fishing capacity (Su et al., 2020).

The Chinese context provides a unique setting to analyze the relationship between enhancing capacity subsidies and fishing activity. Given its fishing history, installed capacity, and the combination of recent policy changes, it allows us to better understand the dynamics of fishers' behavior, revealing their preferences in deciding where to fish as cost structures evolve—whether through direct cost changes associated with the fuel subsidy program or through opportunity costs shaped by the VBP.

2.2.1 Fuel subsidy program

The growth of fishing capacity in China is linked to input subsidy programs (Yang et al., 2017). The Chinese government has long implemented policies aimed at enhancing its fishing capacity (Mallory, 2016). Beginning in the 1980s, the state insulated its economy from international fuel price volatility by subsidizing fuel—purchasing it at high global prices and reselling it domestically at lower rates. However, fiscal deficits in the early 2000s prompted a policy shift in 2006, re-exposing the economy to global fuel prices. In response, the government introduced targeted subsidies for vulnerable sectors such as fisheries, with allocations based on factors such as fuel consumption, gear type, engine power, and vessel size (Englander et al., 2025).

In 2015, facing international pressure to address the ecological harm linked to fisheries subsidies, and in alignment with its *Ecological Civilization* agenda, the Chinese Communist Party (CCP) reformed the program (Su et al., 2020). The previous design had fueled high expenditures, increased fuel use, and exacerbated overfishing (Cao et al., 2017). The reform introduced a classification system for vessels over 12 meters in length, dividing them into 12 groups by size. Each vessel was then assigned a subsidy coefficient based on gear type, with less environmentally friendly gear receiving lower subsidy weights. Furthermore, the baseline fuel price standard was reduced by 18% per year. The reform was implemented starting in 2016. At the beginning of each year, the Fishery Department adjusts the fuel subsidy payment standards based on the previous year's diesel prices, and then proceeds with the processing and disbursement of payments to fishers (Wang et al., 2023).

The reform introduced multiple discontinuities in subsidy allocation: a general one based on gross tonnage and engine power (Englander et al., 2025), applicable to all vessel operations, and a more specific one based on vessel size for operations classified as harmful (e.g., trawlers) (Wang et al., 2023). In this study, we focus on trawler vessels, which are considered one of the most environmentally damaging fishing methods (Buhl-Mortensen and Buhl-Mortensen, 2018). Accordingly, in terms of vessel size, fuel subsidy provisions decreased across all vessels; however, the reduction was more pronounced for vessels under 30 meters in length compared

to those over 30 meters. Smaller vessels experienced an approximately 15.8% reduction in fuel subsidy payments; prior to the reform, there were no differences between the two groups. Figure 2 shows that, prior to the implementation of the 2016 reform, fuel subsidy provision was similar across both types of vessels. After the reform, smaller vessels received significantly lower subsidies³. This discontinuity, in particular, provides the methodological foundation necessary for our analysis. Figure A1 in the appendix shows this relationship in terms of subsidy payments before and after the reform.

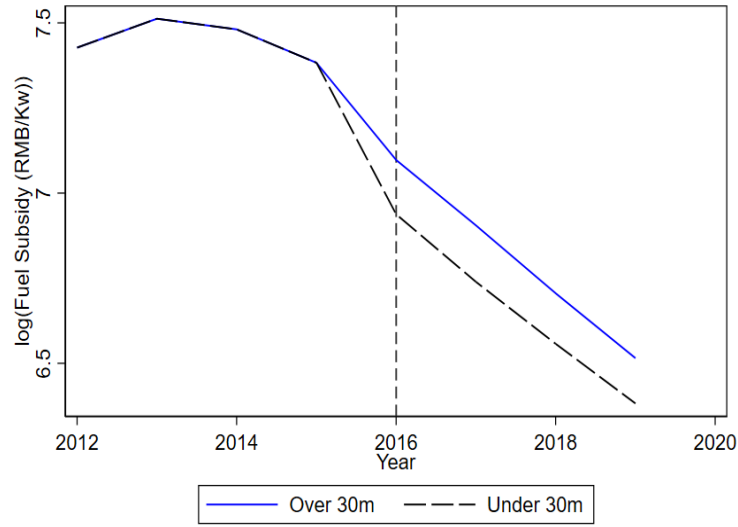


Figure 2. Fuel subsidy payments. Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database. The size of the vessels is measured in meters. The vertical dashed line indicates the year 2016, when the reform was implemented.

2.2.2 Vessel buyback program

The management context of the fishing sector in China is complex, involving the integration of various programs aimed at reconciling economic growth with sustainable sector development. Alongside the subsidy program, since 2002, the VBP has been implemented. This program provides funds to fishers willing to exit the fishing market and transition to non-fishing activities, coupled with the relinquishment of the power quota granted for vessel operation ([Graff Zivin and Mullins, 2015](#)).

The purpose of the VBP is to remove from the fishing industry those vessels that exert the greatest impact on the ecosystem—such as trawlers—by discouraging their continued participation in fishing and increasing their opportunity costs of remaining in the sector. Between 2002 and 2014, it enabled a 31% reduction in the number of commercial fishing

³Fuel subsidies are expressed as the annual payment in Chinese Yuan (RMB) per Power (Kw).

vessels (Cao et al., 2017). This program is grounded in the Quota Management Program for fishing power, which operates in conjunction with the Licensing System. Under this system, each vessel must annually undergo registration, inspection, and management with the Ministry of Agriculture to obtain a fishing license (Wang et al., 2023). The Licensing System was introduced as a regulatory mechanism to curb the disproportionate growth in the number of fishing vessels by requiring that new vessels be constructed only through the acquisition of licenses from decommissioned ones, thereby creating a market for licenses and, together with related policy instruments, aiming to achieve what has been termed *negative growth* in fleet capacity (Su et al., 2020).

Despite the CCP’s efforts to control the environmental impacts of the fishing fleet size, these measures had limited effects due to the coexistence and poor design of the subsidy program, which encouraged fishers to remain in the market (Cao et al., 2017). This situation changed with the 2016 reform. Savings from the redesign of the subsidy program allowed for an increase in the bonuses granted through the VBP. This increase was particularly significant in Zhejiang Province, which, as part of its own objective to reduce the size of its fishing fleet, raised the bonuses beyond the national adjustment. These measures contributed to reducing the fleet’s overall capacity through subsidy cuts (Wang et al., 2023).

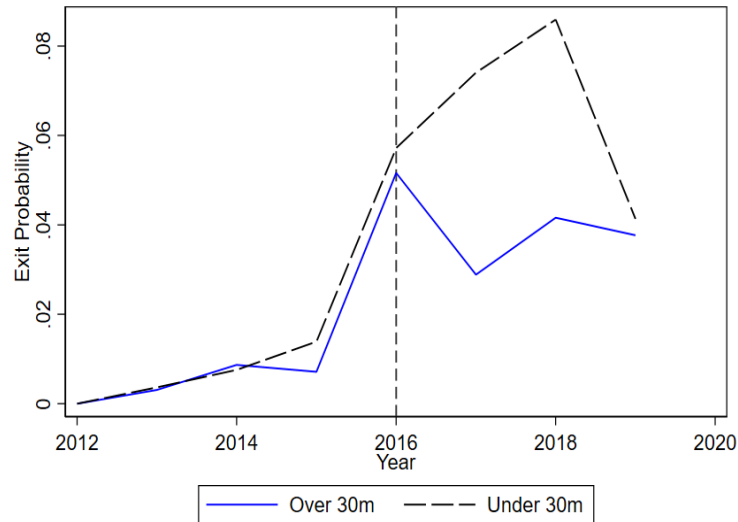


Figure 3. Exit probability. Authors, using information from Global Fishing Watch and Wang et al. (2023) database. The size of the vessels is measured in meters (m). The vertical dashed line indicates the year 2016, when the reform was implemented.

Given the structure and combination of the programs, trawler-type vessels were immediately affected, as they were the primary target due to their environmentally harmful nature (Graff Zivin and Mullins, 2015). Moreover, the interaction between the subsidy reform and

the VBP provides a unique opportunity to analyze how changes in cost structure may reduce incentives to engage in certain IUU fishing activities. Figure 3 illustrates the exit rates from the fishing sector under the VBP, comparing vessels smaller than 30 meters with those larger than 30 meters. A 2.5% higher exit rate is observed among smaller vessels, which experienced a 15.8% greater reductions in subsidies following the reform relative to larger vessels. This suggests that fishers considered their cost structures and expected benefits when deciding whether or not to remain in the sector.

The Chinese case presents a valuable opportunity to examine how fishing incentives and decision making are shaped by changes in subsidy provision and opportunity costs, the latter captured by the offer of exit bonuses from the fishing sector. This context also allows an assessment of how unauthorized fishing dynamics are influenced, especially in the geopolitically contested waters of the East China Sea. Removing financial incentives, such as fuel subsidies, is expected to contribute to reductions in both fishing overcapacity (Wang et al., 2023) and resource overexploitation (Englander et al., 2025). However, the relationship between subsidy reform and IUU fishing is not straightforward. It is therefore essential to understand how vessels respond to changes in their cost structures and potentially to changes in their expectations.

3 The Model

Consider a fishery with $n \in \mathbb{N}$ vessels, indexed by $i = 1, \dots, n$, where n is large. Vessels are either small (s) or large (l), with type $t_i \in \{s, l\}$. Let $S_i \in \mathbb{R}_+$ denote the subsidy received by vessel i . Before the reform, subsidies depend on vessel type, reflecting differences in capacity and fuel use. Reform r reduces subsidies unevenly across vessel types, such that

$$S_i^r = (1 - \lambda_{t_i})S_i. \quad (1)$$

Where $0 < \lambda_l < \lambda_s < 1$, each vessel chooses effort to maximize payoffs. Effort is allocated between authorized (e_{a_i}) and unauthorized (e_{na_i}) fishing, so total effort is $e_i = e_{a_i} + e_{na_i} \in \mathbb{R}_+$. Fishing generates benefits through a concave, increasing revenue function $b_i(e_i)$, reflecting decreasing marginal productivity due to operational constraints such as limited gear, spatial coverage, or handling time. Fishing also entails costs. Let c_{a_i} and c_{na_i} denote the constant marginal costs of authorized and unauthorized fishing, respectively⁴. Payoffs depend on aggregate effort $E_{-i} = \sum_{j \neq i} e_j$ of other vessels, which reduces individual catch opportunities and generates a negative externality $N_i(E_{-i}) \in \mathbb{R}_+$.

Assuming payoff symmetry across vessels⁵, payoffs are defined as

⁴This assumption holds because fishing effort is measured in standardized activity units (e.g., fishing trips or fishing days), so that each additional unit of effort requires similar inputs such as fuel, labor, and vessel operation.

⁵Although larger vessels may achieve higher profits due to scale or technology, the qualitative results remain

$$\pi_i(e_{a_i}, e_{na_i}; E_{-i}) = b_i(e_{a_i} + e_{na_i}) - (c_{a_i} - S_i)e_{a_i} - c_{na_i}e_{na_i} - N_i(E_{-i}). \quad (2)$$

Maximizing vessel payoffs yields the optimal allocation of effort and the corresponding first-order condition for authorized effort

$$\frac{\partial \pi_i(e_{a_i}, e_{na_i}; E_{-i})}{\partial e_{a_i}} = \frac{\partial b_i(e_{a_i} + e_{na_i})}{\partial e_{a_i}} - (c_{a_i} - S_i) = 0. \quad (3)$$

This condition implies that, at the optimum,

$$\frac{\partial b_i(e_{a_i} + e_{na_i})}{\partial e_{a_i}} = c_{a_i} - S_i. \quad (4)$$

In (4), the subsidy lowers the effective marginal cost of authorized fishing, thereby increasing the incentive to allocate effort to authorized fishing activities. Based on equations (1) and (4), the following proposition characterizes how the subsidy reform affects the allocation of authorized fishing.

Proposition 1. *A reduction in subsidies increases the effective marginal cost of authorized fishing, reducing effort. As small vessels experience proportionally larger cuts, their decline in effort may be more pronounced.*

Proof: See Appendix B1.

Considering this adjustment in authorized fishing, the optimal allocation of unauthorized fishing is determined by the first-order condition

$$\frac{\partial \pi_i(e_{a_i}, e_{na_i}; E_{-i})}{\partial e_{na_i}} = \frac{\partial b_i(e_{a_i} + e_{na_i})}{\partial e_{na_i}} - c_{na_i} = 0. \quad (5)$$

Before the reform, authorized fishing is more profitable than unauthorized fishing due to subsidies, such that

$$c_{a_i} - S_i < c_{na_i}. \quad (6)$$

After the reform, the effective marginal cost of authorized fishing becomes $c_{a_i} - S_i^r$. If the subsidy reduction is sufficiently large, the relative cost advantage of authorized fishing may disappear, leading to the following cost structure

$$c_{a_i} - S_i < c_{na_i} \geq c_{a_i} - S_i^r. \quad (7)$$

Changes in costs may alter vessels' effort allocation across fishing activities.⁶ Based on

unchanged. For simplicity, we consider economically homogeneous vessels.

⁶A flexible interpretation of equation (7) allows for the possibility that the inequality $c_{na_i} < c_{a_i} - S_i^r$ reflects either the actual economic cost structure or the behavioral response of fishers to the subsidy reform. In other words, after the reform, the higher net cost of authorized fishing allows fishers to perceive unauthorized fishing as relatively more attractive. This perception may arise from changes in expected revenue, higher opportunity costs of authorized fishing, or imperfect information about enforcement and profitability. Thus, the inequality

equations (1), (6), and (7), the following proposition characterizes the impact of the subsidy reform on unauthorized fishing.

Proposition 2. *A reduction in subsidies increases the relative cost of authorized fishing, potentially prompting vessels to shift effort toward unauthorized fishing. Small vessels, facing proportionally larger cuts, may respond more strongly.*

Proof: See Appendix B2.

Intuition follows from changes in relative marginal costs. If the subsidy reduction increases the effective cost of authorized fishing above that of unauthorized fishing, vessels may reallocate effort toward unauthorized activities. Assuming payoff symmetry, small vessels face stronger incentives due to larger proportional cuts. However, if unauthorized fishing costs are higher for small vessels ($c_{na_i}^s > c_{na_i}^l$) —e.g., due to limited fuel, shorter time at sea, or higher detection risk— larger vessels may increase unauthorized effort more. Thus, the direction of heterogeneous responses cannot be determined theoretically and must be assessed empirically.

The model also provides implications for the overall level of fishing activity.

Proposition 3. *A reduction in subsidies is expected to lower total fishing effort, with small vessels potentially reducing effort more due to proportionally larger cuts.*

Proof: See Appendix B3.

Although some vessels may increase unauthorized fishing, reductions in authorized fishing are not fully offset because authorized and unauthorized fishing are imperfect substitutes. As a result, total fishing effort is expected to decline. Changes in the cost structure may also affect vessel participation in the fishery.

Proposition 4. *A reduction in subsidies may induce less competitive vessels to exit the fishery, particularly when exit incentives such as VBP are available.*

Proof: See Appendix B4.

Higher operating costs reduce the profitability of fishing, potentially prompting some vessels to exit the fishery. In the Chinese context, this mechanism is particularly relevant because the subsidy reform was implemented alongside a vessel buyback program.

In (2), vessels maximize private profits without internalizing the negative externality from aggregate effort. While this externality does not alter individual best responses, it lowers net benefits. Absent productivity-enhancing technologies, reducing total effort decreases private profits, but may increase social welfare if it sufficiently mitigates the externality. The necessary condition for an increase in total net benefits is

$$\sum_{i=1}^n b_i^r - c_i^r + \sum_{i=1}^n \Delta N_i > \sum_{i=1}^n b_i - c_i. \quad (8)$$

Here, b_i and c_i denote generalized benefits and costs, and $\Delta N_i = N_i(E_{-i}) - N_i(E_{-i}^r) > 0$ represents the reduction in negative externality from lower aggregate effort. Since $b_i^r - c_i^r <$

allows the model to capture both direct cost effects and strategic adjustments in effort allocation under the new subsidy regime.

$b_i - c_i$ for all vessels, welfare improves only if the externality reduction is sufficiently large. Regulators typically lack complete information on vessels' payoffs and externalities. In this context, the authorized effort-to-subsidy ratio, $\eta_i = \frac{e_{a_i}}{S_i}$, provides a practical proxy for subsidy efficiency, measuring authorized fishing per unit of subsidy. An increase in η_i may signal that public expenditure continues to generate substantial fishing pressure despite the reform. These considerations motivate the following proposition.

Proposition 5. *Without complementary management, a subsidy reduction may increase the authorized effort-to-subsidy ratio if effort does not fall proportionally.*

Proof: See Appendix B5.

The efficiency condition defines a threshold determined by the costs of authorized fishing, the size of the subsidy reduction, and the benefits from fishing activity (see Appendix B5). Ignoring these interactions may lead to reforms that yield smaller efficiency gains than expected.

Overall, the model generates several empirical implications. A reform combining differential subsidy reductions with a vessel buyback program is expected to reduce authorized and total effort and may induce exit among less competitive vessels. Higher relative costs of authorized fishing may also incentivize shifts toward unauthorized activities. Additionally, the reform may alter the efficiency with which subsidies translate into fishing pressure, as reflected in the authorized effort-to-subsidy ratio. These insights have policy implications: if subsidy cuts increase unauthorized fishing, complementary monitoring and enforcement may be needed; if they primarily affect subsidy efficiency, additional management instruments may be required. These predictions motivate the empirical analysis in the next section.

4 Methodology

4.1 Data Collection

To examine the relationship between access to fuel subsidies for industrial fishing vessels and fishing activity, we used two main data sources. The first is a newly released database compiled by Wang et al. (2023) that contains individual-level administrative data on fishing vessels registered in the Marine Fishing Vessel Dynamic Management System of the Zhejiang provincial government. This administrative platform comprises five modules corresponding to each section of vessel management activities: engine power quota, vessel name, vessel inspection, vessel registration, and fishing license. The second is data from GFW (GFW), which provides information on the number of fishing hours carried out by each vessel at a resolution of 0.01 degrees per pixel on a daily basis.

The database from Wang et al. (2023) contains reports from 7,592 vessels constructed between 1988 and 2011. A total of 3,354 vessels, representing 44% of the sample, were successfully matched with the GFW data. The resulting database includes fishing activity in

hours per day/pixel for 3,354 trawler vessels from Zhejiang province in China for the period 2012-2019. The database also includes characteristics of the vessels, records of fuel subsidy payments received, and records of exits through the VBP.

The descriptive statistics for the outcome variables are presented in Table 1. On average, vessels spend 1.26 hours/day/pixel on fishing activities (3.67 hours/day/pixel for positive fishing hours). In terms of authorization status, vessels averaged 1.32 hours/day/pixel in authorized fishing and 0.23 hours/day/pixel in unauthorized fishing. For positive fishing hours, the averages were 3.67 and 4.09 hours/day/pixel, respectively. No-positive fishing hours refer to pixels where fishing vessels passed through, but did not engage in any fishing activity. This indicates the transit from one fishing location to another. The statistics presented without zeros refer to the exclusion of records in which only transit occurred, considering only those instances where fishing activity actually took place. Regarding the probability of exit due to the VBP, on average, 2.6% of vessels exited the fishing sector, with smaller vessels showing the highest exit rates (Figure 3).

Table 1: *Summary statistics of the main outcome variables before and after reform.*

	Total		Before reform		After reform	
	Mean	SD	Mean	SD	Mean	SD
Total fishing hours	1.26	3.06	1.38	3.11	1.12	2.99
Total fishing hours (positive values)	3.67	4.29	3.56	4.15	3.86	4.51
Authorized fishing hours	1.32	3.11	1.43	3.15	1.18	3.05
Authorized fishing hours (positive values)	3.67	4.28	3.55	4.14	3.86	4.49
Unauthorized fishing hours	0.23	1.55	0.21	1.49	0.24	1.59
Unauthorized fishing hours (positive values)	4.09	5.23	4.12	5.17	4.07	5.27
Fuel Subsidies (RMB/Kw)	1,378.27	434.52	1,727.65	93.29	949.78	276.4
Authorization status	0.95	0.21	0.96	0.19	0.94	0.24
Exit probability	0.026	0.16	0.009	0.09	0.05	0.21

Source: Authors, using information from [Global Fishing Watch](#), [Sea Around Us](#) and [Wang et al. \(2023\)](#) database. Note: Statistics are presented for vessels between 25 and 35 meters in length. Fishing hours are presented as total hours per pixel/day. Each pixel has a resolution of 0.01 degrees, approximately 1km x 1km. Fuel subsidies are expressed as the annual payment in Chinese Yuan (RMB) per Power (Kw). Authorization status takes the value of 1 if the detected fishing activity carried out by a Chinese vessel occurs in the waters of countries where there is a fishing access agreement, in national waters, or in high sea. Exit probability indicates whether the vessel participates in the VBP.

Table 1 also presents the descriptive statistics before and after the reform. On average, subsidy payments decreased by approximately 45%. In terms of fishing activity, total fishing

effort declined by 18% and authorized fishing by 17.5%. By contrast, unauthorized fishing increased by 14.3%, a change associated with a decline in the authorization rate from 96% to 94%. Likewise, the probability of vessel exit increased significantly. Table A1 in the appendix reports additional descriptive statistics.

Figure 4 illustrates the extent of the fishing activity of the vessels in the database, along the eastern coast of China. For positive fishing hours, vessels fished an average of 3.67 hours/day/pixel in national territory, 4.90 hours/day/pixel in Taiwanese territory, 4.57 hours in the joint regime area between Japan and Korea, 3.70 hours in Disputed Zones, and 3.67 hours in Japanese territory (Table 2).

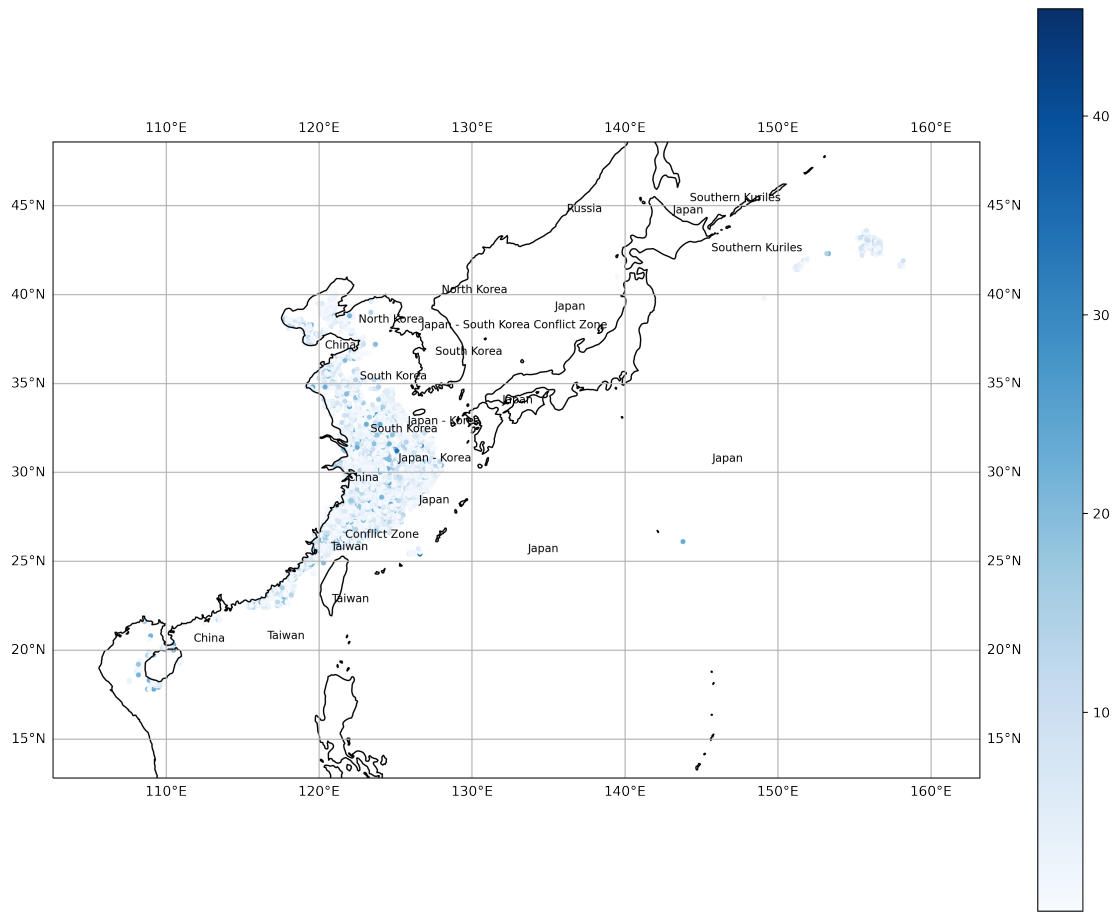


Figure 4. Coverage of Chinese trawler vessel's fishing activity. Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database. Each pixel has a resolution of 0.01 degrees, approximately 1km x 1km.

Regarding the authorization status of fishing activity, we rely on the Sea Around Us (SAU) database on fishing access agreements. Fishing activity is classified as authorized if a formal agreement exists between the parties. This distinction is necessary because the SAU database includes not only formal agreements but also vessel detection information derived from GFW

data, making it essential to clearly differentiate between the types of access reported. Figure A2 in the appendix shows the distribution of vessel length by type of fishing activity for vessels between 25 and 35 meters in length, both before and after the reform. Following the reform, the distribution shifted toward a greater share of unauthorized fishing. Table A2 in the appendix shows the fishing agreements that China had during the analysis period⁷.

Table 2: *Chinese fishing activity by countries and sovereign.*

Country	Sovereign	Agreements	Fishing Hours > 0		
			Obs	Mean	SD
China	China	-	1,466,122	3.67	4.28
South Korea	South Korea	Yes	1,107	3.95	4.87
Taiwan	Taiwan	No	2,872	4.90	5.46
Disputed Zone	Disputed	No	2,111	3.70	4.60
Japan - Korea	Joint Regime	No	1,198	4.57	5.87
Japan	Japan	No	521	3.99	4.64

Source: Authors, using information from [Global Fishing Watch](#), [Sea Around Us](#), and [Wang et al. \(2023\)](#) database. Note: Positive fishing hours are presented. The agreements were defined according to the internal fishing access agreements dataset from Sea Around Us. The existence of an agreement is determined by whether a formal agreement was signed between the parties for the years under analysis.

4.2 Empirical Strategy

China’s fuel-subsidy reform created two key discontinuities: one based on gross tonnage and engine power ([Englander et al., 2025](#)), and another tied to vessel size for harmful operations such as trawling ([Wang et al., 2023](#)). Our analysis concentrates on the vessel-size threshold because it aligns with our dataset and the policy’s focus, while the second discontinuity does not clearly separate binding from non-binding vessels.

This redesign motivates the implementation of a regression discontinuity model, notably, vessels just over 30 meters in length received approximately 19% more in subsidy payments than those just under 30 meters. However, the reform affected not only the margin around the cutoff, but also the continuum of vessels. For some, the subsidy reduction compared to the previous year was much greater, but it decreased for all. Thus, a DiD estimation can capture more information than a regression discontinuity (RD) model, given that RD estimates are local and concentrated around the cutoff point. Considering these factors, we employ a DiD model to obtain more generalizable estimates, differentiating the vessel population by size in line with the program’s design, and restricting the sample to vessels between 25 and 35

⁷For agreements with unspecified end year, we assume that the agreement is still valid, which biases the authorization status towards authorized rather than unauthorized, therefore, the results obtained could be interpreted as a lower bound.

meters in length.

The proposed model aims to estimate the effect of differential reduction in subsidy allocation on fishing activity and the number of unauthorized fishing hours, using the following specification

$$y_{izt} = \alpha_1 D_i + \alpha_2 Post_{t_y} + \beta D_i \times Post_{t_y} + \gamma_i + \phi_z + \rho_t + \epsilon_{izt}. \quad (9)$$

Where y is the unauthorized fishing efforts for vessel i at pixel z on the day t . The variable $D_i = \mathbb{1}\{\text{Vessel Length} \geq 30\}$ indicates whether the vessel has a length greater than 30 meters. This is a threshold chosen based on the evaluation of subsidy access distribution by vessel size (Figure 2). $Post_{t_y}$ takes the value of 1 from year $t_y = 2016$, when the subsidy program reform was implemented, and 0 otherwise. The variables γ_i , ϕ_z y ρ_t represent vessel fixed effects, pixel fixed effects, and time fixed effects, respectively. The idiosyncratic error is represented by ϵ_{izt} . The coefficient of interest is β which captures the effect of the reform to the fuel subsidy program by large vessels.

To avoid potential bias from manipulation of the treatment assignment rule, the database consists of vessels built before 2012. Given this, and considering that D_i is a characteristic that does not vary over time in the analysis period, there is no reason to believe that D_i is correlated with unobserved factors that determine the outcome variable. Therefore, it is not necessary to include additional covariates.

Taking into account the distribution of vessels by size, the regressions are estimated for vessels around the 30-meter threshold. As the main estimation, the regressions are conducted for vessels between 25 and 35 meters.

Based on equation (9), we made the following change: similar to Wang et al. (2023), we replaced $\beta D_i \times Post_{t_y}$ with $\beta_j D_i \times Post_{t_y}^j$ for $j = 2012, \dots, 2019$, where $Post_{t_y}^j = \mathbb{1}\{t = j\}$. This approach allows us to estimate the treatment effect over time by conducting an event study that captures the impact of the subsidy reduction resulting from the program reform. We use the following specification

$$y_{izt} = \alpha_1 D_i + \alpha_2 Post_{t_y} + \sum_{j=2013}^{2019} \beta_j D_i \times Post_{t_y}^j + \gamma_i + \phi_z + \rho_t + \epsilon_{izt}. \quad (10)$$

To ensure correct estimation of the causal effect of the change in subsidy access on unauthorized fishing activity using the DiD method, it is necessary to satisfy the following assumptions: (i) *No anticipation*. This refers to the assumption that fishers did not have the opportunity to adjust vessel characteristics to take advantage of the treatment assignment rule. To prevent any bias, only vessels built before 2012 are considered. (ii) *Parallel trends*. This assumption presupposes that the fishing behavior of vessels that had access to more subsidies and those that had access to fewer subsidies follows parallel trends before the reform. (iii) *No interference between units (SUTVA)*. It is possible that this assumption is

violated, which calls for a cautious interpretation of the results. The estimation captures the difference between the impact on the treatment group and the control group; however, because the control group is also affected by the treatment, the results differ from what would have been observed if the treatment group had not been exposed to the reform at all. While this approach still allows us to identify the relative differential effect of the reform across the defined groups, the results should be interpreted as a *relative effect* rather than the absolute impact of the treatment compared to a purely untreated counterfactual scenario.

5 Results

In this section, we present results showing how fishers adjust their decisions, reflected in both the number of hours spent fishing and the locations where fishing takes place, in response to reductions in fuel subsidy payments and to shifts in opportunity costs as measured by the VBP. Regarding location choices, we distinguish between fishing activity conducted in authorized areas (national waters, the high seas, and the national waters of countries with which valid access agreements exist) and unauthorized fishing, which refers to fishing undertaken in the national waters of countries without such agreements or in areas currently designated as disputed.

5.1 Effects of subsidy reform on total effort

We find that the reform reduced the total average of subsidy payments by approximately 69% (Figure A3 in the appendix). However, it resulted in larger vessels receiving 15.8% more in fuel subsidy payments relative to smaller vessels (Table A3 in the appendix). This redistribution led to a decrease in overall fishing effort, with a more pronounced reduction among smaller vessels—driven by their relatively lower post-reform subsidies (Figure 5). These results suggest that the reduced availability of subsidies for smaller vessels altered their cost structure, further compounded by changes in opportunity costs driven by the VBP.

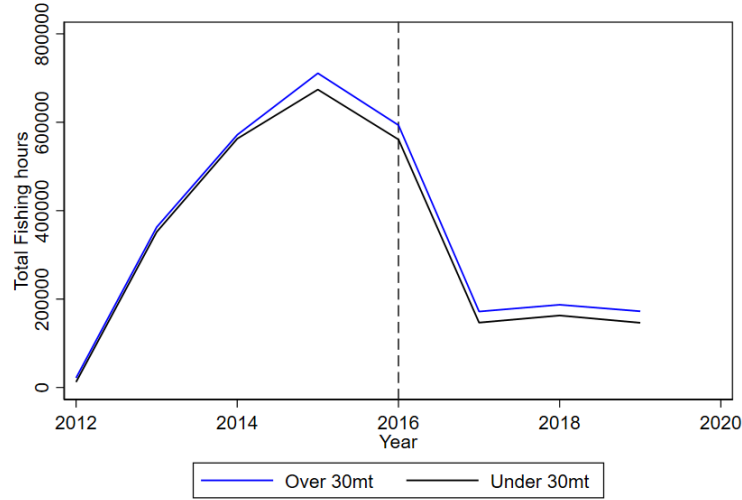


Figure 5: Total fishing efforts by treatment. Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database.

The adjustment in subsidy allocation, combined with the VBP program, made smaller vessels 2.5% more likely to exit the fishing sector than larger ones, and less competitive vessels gradually exited the fishery. This finding aligns with the results reported by [Wang et al. \(2023\)](#). The VBP incentives accelerated the exit of smaller vessels, raising the opportunity cost of staying in the sector. However, it is important to note that the effect was moderate: for each 1% reduction in the subsidy, the probability of exit for smaller vessels increased by only 0.16 percentage points (see Table A3 in the appendix).

5.2 Effects of subsidy reform on authorized and unauthorized fishing

When disaggregating fishing activity by type of authorization, we find that, along with the reduction in authorized activity, there was an increase in unauthorized fishing as a compensatory strategy to offset the loss of competitiveness caused by the lower subsidy. Figure 6 shows that authorized fishing effort began to decline after the reform (vertical dashed line), while unauthorized fishing effort (blue line) increased. In absolute terms, the decline in authorized fishing effort was substantially greater than the increase in unauthorized activity. It should be noted that the share of vessels engaged in unauthorized fishing was less than 5% (Table 1).

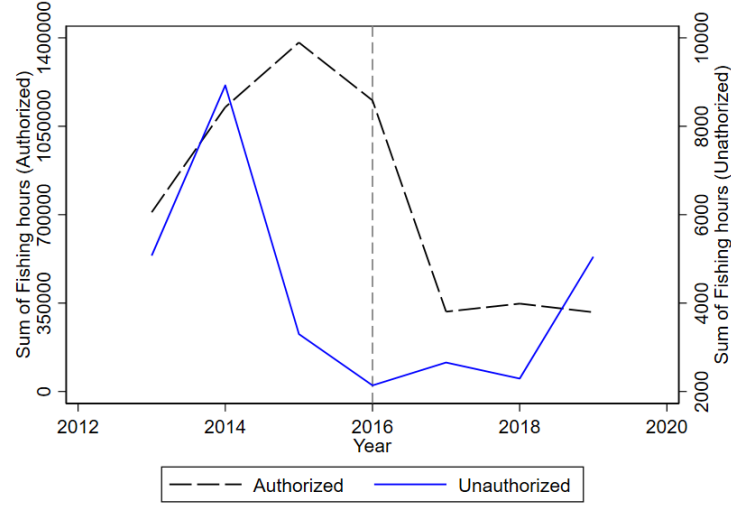


Figure 6: Fishing efforts by authorization status. Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database.

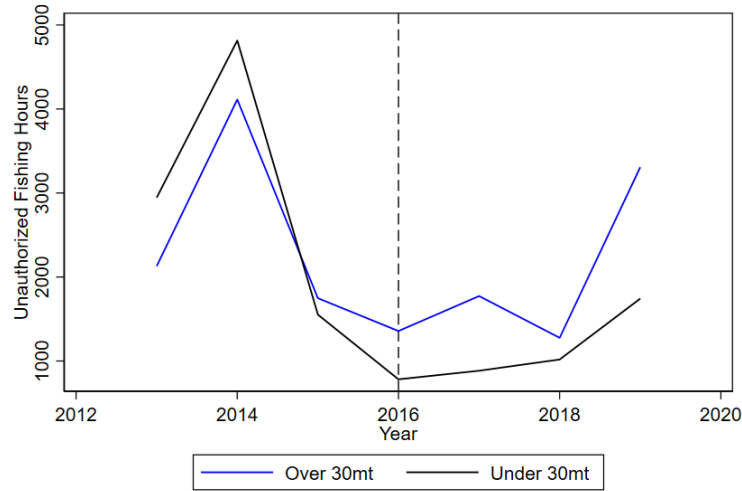


Figure 7. Total unauthorized fishing efforts by vessel length over years. Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database.

We observe that the increase in unauthorized fishing activity was more pronounced among larger vessels (Figure 7). This outcome is expected, as larger vessels have greater physical capacity, allowing them to undertake longer and more distant fishing trips, which gives them a comparative advantage over smaller vessels. Before the reform, smaller vessels engaged in more unauthorized fishing than larger ones, a pattern supported by their access to higher subsidy levels that enhanced their fishing capacity. The reform, by primarily affecting smaller

vessels, discouraged unauthorized fishing among them, while simultaneously creating incentives for larger vessels to expand their activity by taking advantage of the space left by smaller vessels.

Table 3 shows the effect of the reform on unauthorized fishing efforts. It is observed that fishing hours in countries without a fishing access agreement increased by 0.166, meaning that unauthorized fishing after the reform increased by 16.6% for larger vessels. Table A4 in the appendix presents the results of a placebo test for the periods prior to the reform. No significant effects are observed, except for the post-2014 period; however, this effect disappears once fixed effects are included, suggesting that the results in Table 4 are unlikely to be biased by pre-reform factors.

Table 3: *Average treatment effect of fuel subsidy program reform on unauthorized fishing activity*

	Unauthorized fishing hours (<i>log</i>)			
	(1)	(2)	(3)	(4)
1.Treated*1.Post	-0.015 (0.04)	0.189*** (0.06)	0.19*** (0.06)	0.166** (0.07)
Fixed Effect	No	<i>i</i>	<i>i, t</i>	<i>i, t, z</i>
Mean _{<2016} Log(<i>y_{izt}</i>)	1.03	1.03	1.03	1.03
N Treated	4,868	4,814	4,813	4,764
N Control	3,968	3,934	3,932	3,913
Observations	8,836	8,748	8,745	8,677

Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database. Note: * p<.10, * p<.05, ** p<.01. The outcome variables are expressed in log. Vessel, year, month and pixel Fixed Effect are included. Robust standard deviation in parenthesis. The mean represents the average of the outcome variable before the reform for the control group.

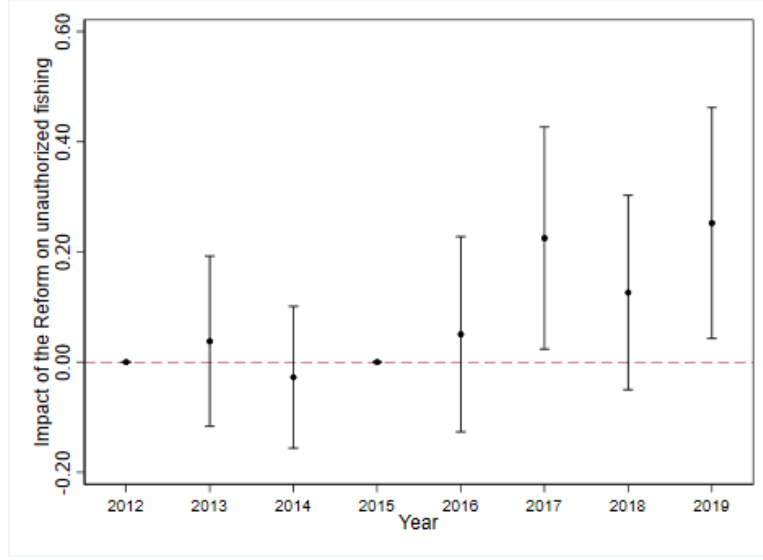


Figure 8: Event study of fuel subsidy program reform effect on unauthorized fishing hours. Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database. Note: The vertical bars indicate the confidence intervals at 10%. The figure presents the results of the estimates based on equation (10).

The effect of the reform on unauthorized fishing activity remained consistent over time and intensified in later years, when the reduction in subsidies became smaller. Figure 8 presents the estimates from equation (10) relative to the year of the reform (event study). It can be observed that the number of unauthorized fishing hours increased from the first year, but it became significant from the year 2017 onwards, remaining so in the year 2019, before the pandemic. Figures A4 and A5 in the appendix present unauthorized fishing hours by destination countries of Chinese fleets (Taiwan, Japan, and Disputed Zones) as well as the event study results. The findings show that the increase in unauthorized fishing activity occurs primarily in Disputed Zones and in Taiwan waters.

5.3 Effects of subsidy reform on authorized effort-to-subsidy ratio

Finally, we observe that as the subsidy amount decreased, smaller vessels increased their fishing effort per unit of subsidy. In other words, after the reform these vessels raised their fishing effort per RMB/kW of subsidy received—both for authorized and unauthorized fishing. Figure 9 illustrates the evolution of the ratio between fishing hours and fuel subsidies by vessel size. These results also allow for a second interpretation regarding the effectiveness of the reform in shaping fishers' decisions. The reduction in fishing effort was not directly proportional to the reduction in subsidies; rather, as subsidies declined, vessels reported relatively higher levels of effort compared to the amount of subsidy received.

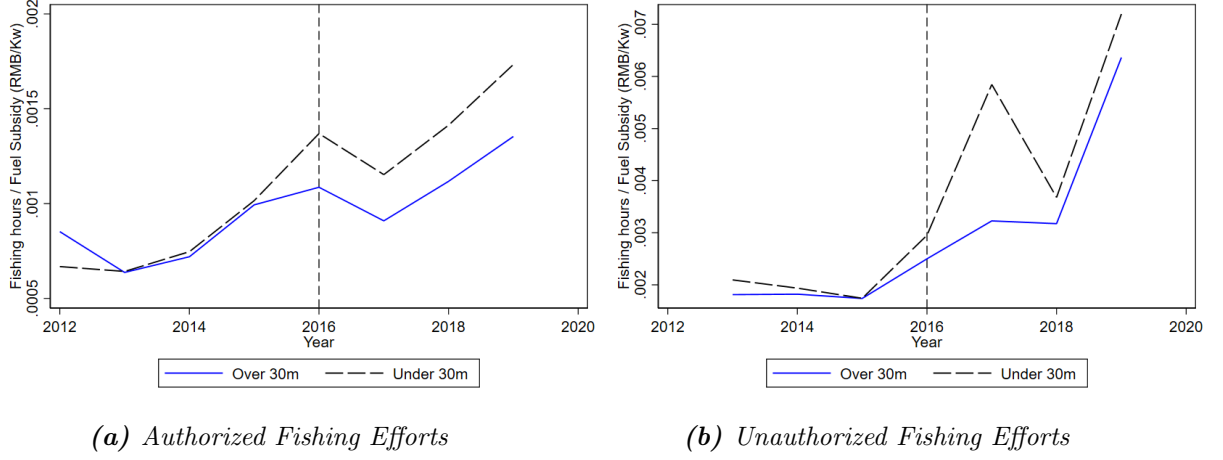


Figure 9. The relationship between fishing efforts and fuel subsidy payments before and after reform by vessel length. Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database.

Specifically for unauthorized fishing, the post-reform efficiency adjustment was observed among both smaller and larger vessels, supporting our hypothesis that the increase in fishing hours was driven by fishers' decisions to compensate for the subsidies they no longer received. This behavior explains the stabilization of authorized fishing effort starting in 2017 (Figure 5) and the rise in unauthorized activity from 2018 onward (Figure 7), even as subsidies continued to decline (Figure A3 in the appendix).

6 Discussion and Concluding Remarks

This paper examines how state-provided economic incentives, specifically fuel subsidies, shape fishing practices by Chinese industrial vessels. In this paper, we propose a theoretical model together with a causal inference model, which allows us to understand and estimate the impact of differential access to subsidies on the decisions of industrial fishers to engage in fishing activities in international waters of countries where the Chinese government lacks a fishing access agreement, that is, to engage in unauthorized fishing activity. A distinguishing aspect of this study is its focus on unauthorized fishing as defined by the [FAO \(2024\)](#), namely, fishing activities carried out in countries without formal access agreements. By separating these from legal operations, we offer a more precise assessment of how subsidies affect compliance with international norms.

Our results suggest that the subsidy reform significantly reduced overall fishing effort, in line with the government's objective of lowering installed capacity through the combined effects of the VBP and the progressive reduction of capacity-enhancing subsidies. Our theoretical model accurately predicts that the differential reduction of subsidies between smaller and larger vessels also generated heterogeneous behavioral responses. Larger vessels, having

received higher subsidy levels compared to smaller ones, reduced their fishing effort to a lesser extent—an outcome explained by the differential shift in cost structures across vessel sizes. This reduction was primarily driven by authorized fishing activity, over which government policy exerts a more direct influence.

With respect to unauthorized fishing, the results show that the reform led to a decline in unauthorized activity among smaller vessels relative to pre-reform levels. However, as subsidy reductions deepened, smaller vessels that remained in the sector—i.e., the more competitive ones not participating in the VBP—intensified their fishing effort relative to the subsidies they received. This indicates that these vessels sought to compensate for their loss of competitiveness, caused by higher relative costs, by increasing fishing activity. In contrast, larger vessels increased their unauthorized fishing following the reform, explained both by the decline in activity among smaller vessels and by their greater installed capacity combined with their relatively higher access to subsidies (Englander et al., 2025), all in an effort to offset rising costs.

In terms of the effort–subsidy relationship, the model strongly predicts that smaller vessels increase this ratio in both authorized and unauthorized fishing. This suggests, in terms of policy effectiveness, that subsidy reductions do not translate into a proportional decline in fishing effort. Instead, fishers’ behavior reveals a compensatory response, whereby subsidies are used more intensively to mitigate the loss of competitiveness. This finding contributes to the broader debate on the implications of input subsidies (Sumaila et al., 2021), which have been shown to threaten ecosystem sustainability by increasing fishing effort (Finley, 2017), enabling longer trips (Englander et al., 2025), and, as demonstrated here, intensifying IUU fishing.

Our findings contribute to related literature in several ways. First, they provide empirical support for the argument that public subsidies can facilitate activities inconsistent with international sustainability goals, especially when they reduce the private cost of long-distance fishing. While previous work has shown that subsidies can distort incentives (Sumaila et al., 2010b, Mallory, 2013). Second, we provide quantitative evidence that vessels receiving greater subsidies increase their presence in foreign waters where China lacks fishing access agreements, with marked increases in Disputed Zones, including Taiwanese waters. This raises concerns about the geopolitical consequences of domestic support programs (Midford and Østhagen, 2024, Zhang, 2016).

This paper contributes to a deeper understanding of how domestic instruments can produce externalities beyond national jurisdiction by documenting the link between national subsidy policies and transboundary fishing behaviour. These results underscore the importance of aligning subsidy reforms not only with domestic objectives, but also with broader international commitments aimed at ensuring equitable and sustainable marine resource use. It is necessary to expand this type of analysis to contexts with weaker enforcement in order to

evaluate how the willingness of fishers to engage in unauthorized fishing activities changes as enforcement decreases. Broadening the scope to include varying institutional and regulatory settings could offer valuable insights into the interplay between enforcement intensity, cost structures, and IUU fishing behavior.

This study has several limitations. First, the main outcome variable, fishing effort, relies on Automatic Identification System (AIS) data as processed by GFW. While this dataset enables consistent global monitoring, it only captures vessels that broadcast AIS signals, potentially excluding those that deliberately disable transmitters to avoid detection. Although GFW applies cross-referencing techniques to mitigate manipulation of location data ([Paolo et al., 2024](#)), unobserved activity remains a concern. Second, the dataset from [Wang et al. \(2023\)](#) is based on official subsidy recipients, which by design excludes vessels not formally compliant with Chinese regulations. Consequently, our estimates likely understate the full extent of unauthorized activity, and may not capture behaviour by habitual violators.

Further research could strengthen these findings along several lines. Integrating non-AIS satellite imagery or port inspection data could help identify vessels operating without digital traceability. Survey evidence from fishing communities may also provide insight into fishers' perceptions of enforcement and how these shape risk tolerance. Finally, comparative studies involving other major distant-water fishing nations would help assess whether subsidy-induced reallocation of effort is a broader phenomenon, and could inform international negotiations on subsidy discipline and fisheries governance.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT by OpenAI to improve the clarity and readability of the manuscript. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the publication.

Supplementary data

Supplementary data to this article can be found online at

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Appendix

A Additional Figures and Tables

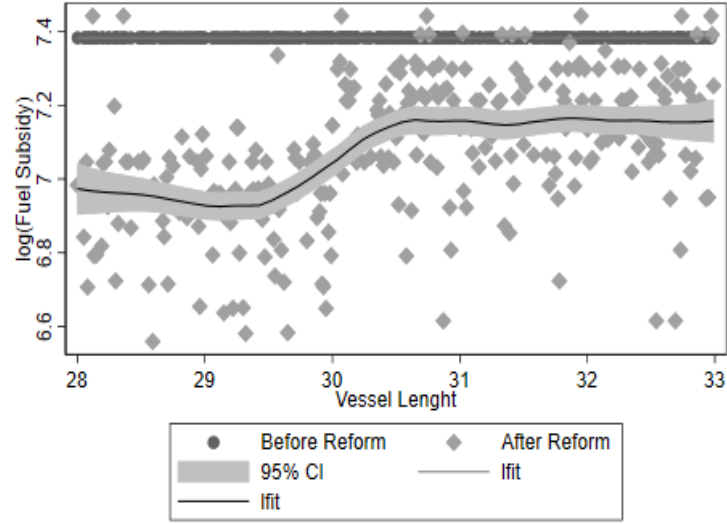


Figure A1. The relationship of fuel subsidy payments per Kw and vessel length in 2015-2016. Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database. Note: Figure shows the total fuel subsidy payments per Kw of engine power after reform.

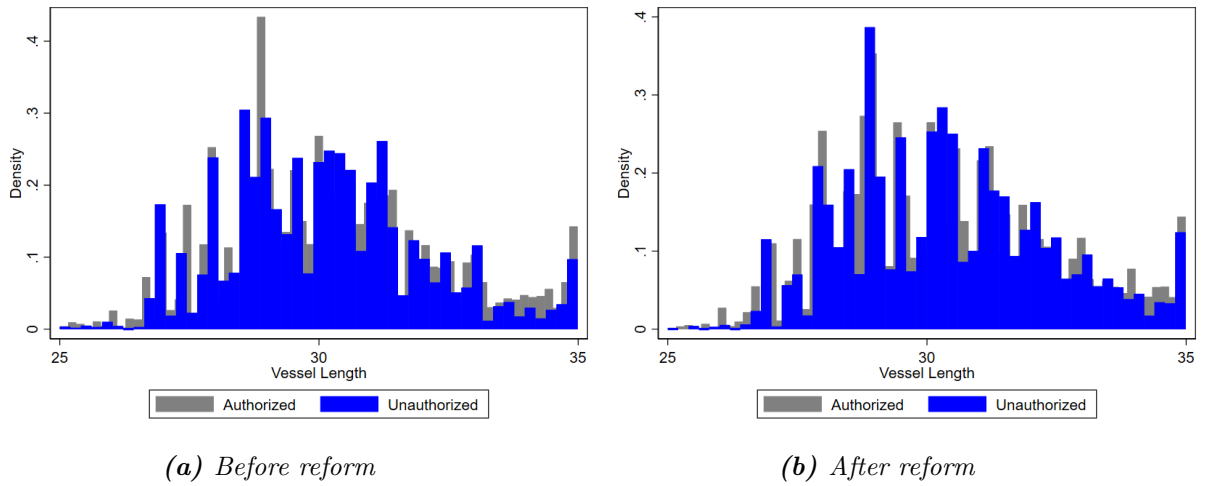


Figure A2: Distribution of vessel length. Source: Authors, using information from [Global Fishing Watch](#), [Sea Around Us](#) and [Wang et al. \(2023\)](#) database. Note: Panel (a) shows the vessel length distribution before the reform, and Panel (b) shows the distribution after the reform.

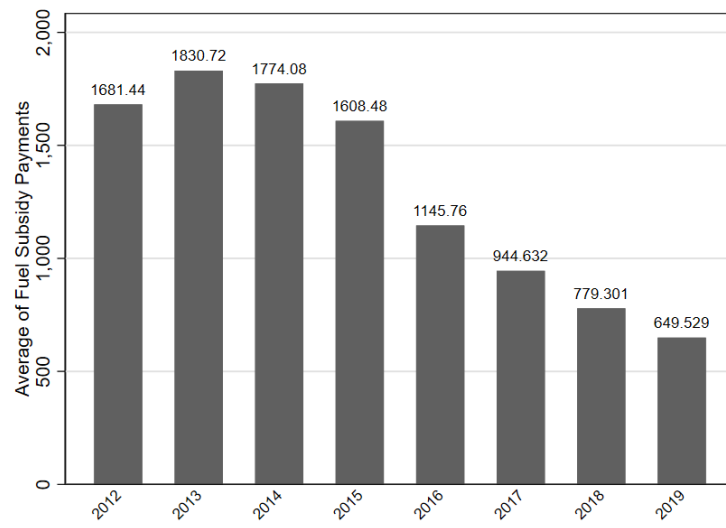
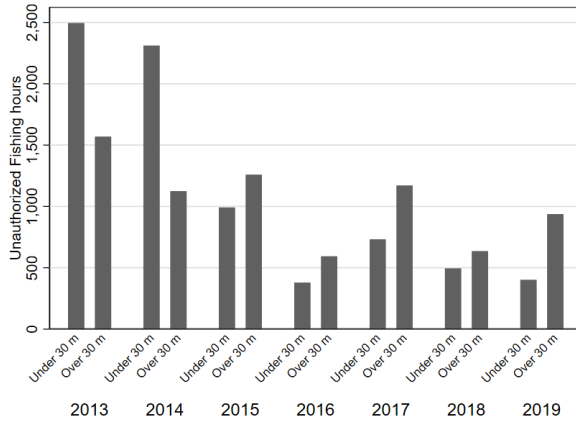
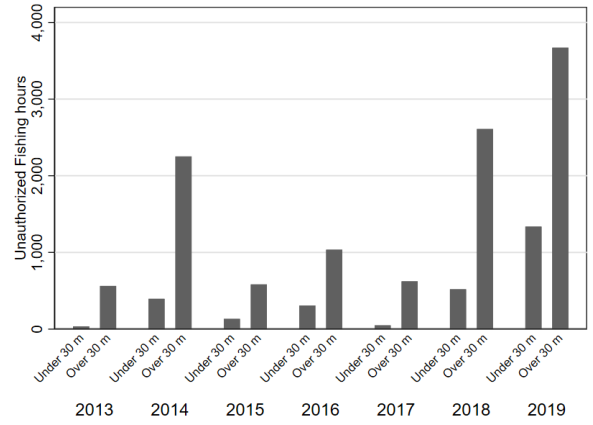


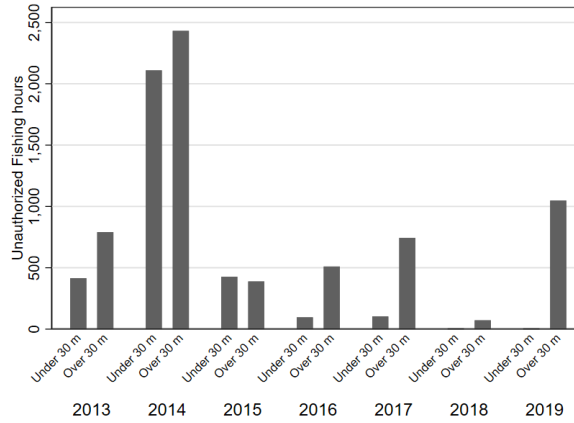
Figure A3: Total average of fuel subsidy payments over years. Source: Authors, using information from [Wang et al. \(2023\)](#) database. Note: Figure shows the average of fuel subsidy payments for vessels between 25 and 35 meters in length.



(a) Taiwan

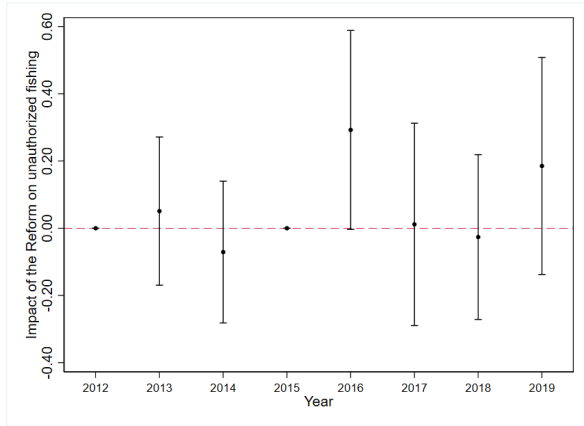


(b) Japan

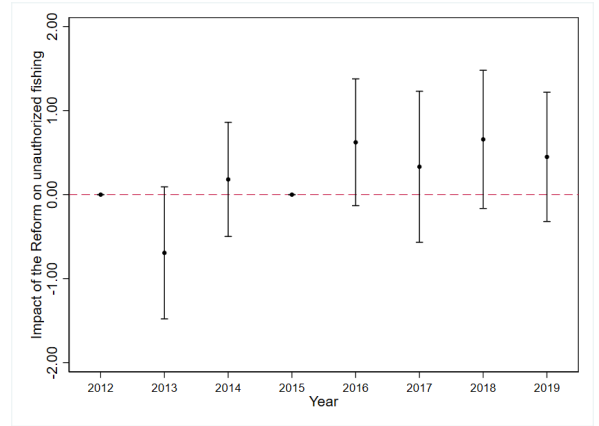


(c) Disputed Zone

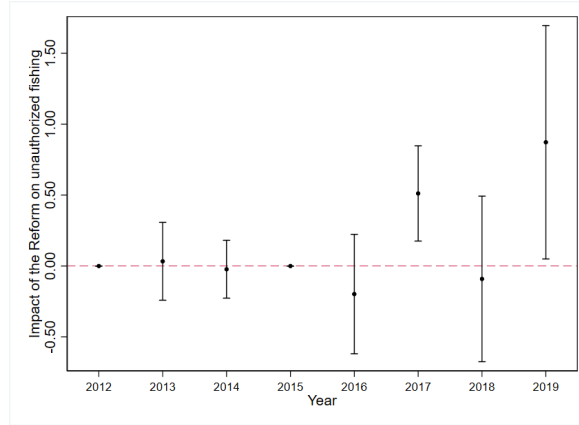
Figure A4. Total unauthorized fishing hours by vessel length over years. Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database. Note: Panel (a) shows the total chinese unauthorized fishing efforts in Taiwan, panel (b) shows the unauthorized fishing hours in Japan, and panel (c) shows the activity in Disputed Zones.



(a) Taiwan



(b) Japan



(c) Disputed Zone

Figure A5: Event study of fuel subsidy program reform effect on unauthorized fishing hours by countries. Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database. Note: Note: Panel (a) shows the effects in Taiwan, panel (b) shows the effect in Japan, and panel (c) shows the effects in Disputed Zones. The vertical bars indicate the confidence intervals at 10%. The figure presents the results of the estimates based on equation (10) for each country.

Table A1: Summary statistics of the main outcome variables.

	Min	25%	50%	Mean	75%	Max	Obs
Total fishing hours	0	0	0	1.25	1.03	45.39	5,074,669
Total fishing hours (without 0)	0.00	0.93	2.07	3.63	4.57	45.39	1,752,117
Authorized fishing hours	0	0	0	1.31	1.15	45.39	4,827,774
Authorized fishing hours (without 0)	0.00	0.93	2.07	3.63	4.57	45.39	1,737,173
Unauthorized fishing hours	0	0	0	0.25	0	39.06	246,895
Unauthorized fishing hours (without 0)	0.00	0.70	2.03	4.16	5.30	39.06	14,944
Fuel subsidies (RMB/Kw)	331	958.75	1608.48	1375.27	1774.08	1830.72	5,074,669
Authorization status	0	-	-	95.1%	-	1	5,074,669
Exit Probability	0	-	-	2.25%	-	1	5,074,669

Source: Authors, using information from [Global Fishing Watch](#), [Sea Around Us](#) and [Wang et al. \(2023\)](#) database. Note: Statistics are presented for vessels between 25 and 35 meters in length. Fishing hours are presented as total hours per pixel/day. Each pixel has a resolution of 0.01 degrees, approximately 1km x 1km. Fuel subsidies are expressed as the annual payment in Chinese Yuan (RMB) per Power (Kw). Authorization status takes the value of 1 if the detected fishing activity carried out by a Chinese vessel occurs in the waters of countries where there is a fishing access agreement, in national waters, or in high sea. Exit probability indicates whether the vessel participates in the VBP.

Table A2: *Chinese internal fishing access agreements.*

Country Owned	Start year	End year
Angola	2016	9999
Argentina	2020	2020
Benin	2015	9999
Cameroon	2018	9999
Côte d'Ivoire	2015	9999
Equatorial Guinea	2018	9999
Fiji	2018	2021
French Polynesia	2014	2019
Ghana	2019	9999
Guinea	2018	9999
Guinea-Bissau	2017	9999
Guyana	1996	9999
Liberia	2018	9999
Madagascar	2019	2021
Micronesia (Federated States of)	2020	2020
Mozambique	2018	2019
Nauru	2020	2020
Nigeria	2018	9999
Oman	2018	2019
Palau	2018	2019
Papua New Guinea	2016	2019
Peru	2010	2018
Senegal	2017	9999
Seychelles	2017	9999
Sierra Leone	2018	9999
Singapore	2017	2019
Somalia	2015	2019
Suriname	2018	2019
South Korea	2001	2019
Togo	2013	9999
Tuvalu	2020	2020
Uruguay	2018	2019
Vanuatu	2020	2020
Viet Nam	2017	2019
Angola	2016	9999
Argentina	2020	2020
Benin	2015	9999
Cameroon	2018	9999
Côte d'Ivoire	2015	9999
Equatorial Guinea	2018	9999
Fiji	2018	2021
Ghana	2019	9999
Guinea	2018	9999
Guinea-Bissau	2017	9999
Guyana	1996	9999
Liberia	2018	9999
Madagascar	2019	2021
Micronesia (Federated States of)	2020	2020
Nauru	2020	2020
Nigeria	2018	9999
Peru	2016	9999
Senegal	2017	9999
Seychelles	2017	9999
Sierra Leone	2018	9999
Togo	2013	9999
Tuvalu	2020	2020
Vanuatu	2020	2020

Source: Internal Fishing Access Agreements - [Sea Around Us](#). Note: Country owned relates the countries with which China has had fishing access agreements. Values of 9999 in End year indicate that there is no specified termination date for the agreement.

Table A3: Average treatment effect of fuel subsidy program reform on fuel subsidy access and exit probability.

	Fuel subsidy (<i>log</i>)		Exit Probability	
	(1)	(2)	(3)	(4)
1.Treated*1.Post	0.141*** (0.0004)	0.158*** (0.0003)	-0.018*** (0.0003)	-0.025*** (0.0004)
Fixed Effect	No	Yes	No	Yes
Mean _{<2016} Log(<i>y_{izt}</i>)	7.45	7.45	0.011	0.011
N Treated	2,274,390	2,164,951	2,274,390	2,164,951
N Control	2,026,315	1,937,387	2,026,315	1,937,387
Observations	4,300,705	4,102,338	4,300,705	4,102,338

Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database. Note: * p<.10, * p<.05, ** p<.01. The outcome variables in columns (1)-(2) are expressed in log. Vessel, year, month and pixel controls are included. Robust standard deviation in parenthesis. The mean represents the average of the outcome variable before the reform for the control group.

Table A4: Pre-Period placebo test.

	Unauthorized fishing hours (<i>log</i>)			
	Post > 2014		Post > 2015	
	(1)	(2)	(3)	(4)
1.Treated*1.Post	0.097* (0.06)	-0.095 (0.08)	0.091 (0.06)	0.102 (0.084)
Fixed Effect	No	Yes	No	Yes
N Treated	2,544	2,500	2,544	2,500
N Control	2,723	2,662	2,723	2,662
Observations	5.267	5.162	5.267	5.162

Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database. Note: * p<.10, * p<.05, ** p<.01. The outcome variables are expressed in log. Vessel, year, month and pixel Fixed Effect are included. The estimators are calculated by replacing the value of *post* with the years 2014, and 2015, excluding the values after 2016, which are the actual treatment years. Robust standard deviation in parenthesis.

B Proofs

B.1 Proof of Proposition 1

Since $S_i^r < S_i$, then $\left. \frac{\partial b_i(e_{a_i} + e_{na_i})}{\partial e_{a_i}} \right|_{e_{a_i}} < \left. \frac{\partial b_i(e_{a_i} + e_{na_i})}{\partial e_{a_i}} \right|_{e_{na_i}}$, implying that, due to concavity, higher marginal benefits denote lower efforts. Consequently, effort after the reform is lower than before the reform, $e_{a_i}^r < e_{a_i}$. Since $\lambda_l < \lambda_s$, then subsidies received after the reform

by small vessels are lower than subsidies received by large vessels, $S_i^{r,s} < S_i^{r,l}$, implying that $\left. \frac{\partial b_i(e_{a_i} + e_{na_i})}{\partial e_{a_i}} \right|_{e_{a_i}^{r,l}} < \left. \frac{\partial b_i(e_{a_i} + e_{na_i})}{\partial e_{a_i}} \right|_{e_{a_i}^{r,s}}$, that is, $e_{a_i}^{r,s} < e_{a_i}^{r,l}$. $\square$

B.2 Proof of Proposition 2

The effect of the subsidy reform on unauthorized efforts is captured by

$$\frac{\partial b_i(e_{a_i} + e_{na_i})}{\partial e_{i \in \{a_i, na_i\}}} = \min\{c_{a_i} - S_i^r, c_{na_i}\} \quad (11)$$

If $c_{a_i} - S_i < c_{na_i} < c_{a_i} - S_i^r$ holds, then $e_{na_i}^r > e_{na_i}$, given that $e_{a_i}^r < e_{a_i}$. since $\lambda_l < \lambda_s$, then subsidies received after the reform by small vessels are lower than subsidies received by large vessels, $S_i^{r,s} < S_i^{r,l}$, implying that $c_{a_i} - S_i^{r,l} < c_{a_i} - S_i^{r,s}$, that is, $e_{na_i}^{r,s} > e_{na_i}^{r,l}$. $\square$

B.3 Proof of Proposition 3

Let $E_T = \sum_{i=1}^n e_i$ denote the total effort exerted by the fishing community, where $e_i = e_{a_i} + e_{na_i}$. Proposition 1 states that $e_{a_i}^r < e_{a_i}$ for all i , such that $E_a^r = \sum_{i=1}^n e_{a_i}^r < E_a = \sum_{i=1}^n e_{a_i}$. Since, $e_{a_i}^{r,s} < e_{a_i}^{r,l}$, then $E_a^{r,s} < E_a^{r,l}$. E_a denotes total authorized effort before the reform and E_a^r total authorized effort after the reform. Proposition 2 states that $e_{na_i}^r > e_{na_i}$ for all i , such that $E_{na}^r = \sum_{i=1}^n e_{na_i}^r > E_{na} = \sum_{i=1}^n e_{na_i}$. E_{na} denotes total unauthorized effort before the reform and E_{na}^r total unauthorized effort after the reform. Since $e_{na_i}^{r,s} > e_{na_i}^{r,l}$, then $E_{na_i}^{r,s} > E_{na_i}^{r,l}$. The concavity of the benefit function and the cost structure, $c_{a_i} - S_i < c_{na_i} < c_{a_i} - S_i^r$, imply that authorized and unauthorized efforts are imperfect substitutes, such that $e_{a_i}^r + e_{na_i}^r < e_{a_i} + e_{na_i}$. If $c_{a_i} - S_i < c_{na_i}$, then $e_{na_i} = 0$ and therefore $e_{a_i}^r + e_{na_i}^r < e_{a_i}$ for all i . This leads to $(E_T^r = E_a^r + E_{na}^r) < (E_T = E_a)$. $\square$

B.4 Proof of Proposition 4

A necessary condition to exit the fishery is a loss of competitiveness, represented by a decrease in payoff below a certain threshold level, denoted by $B \in \mathbb{R}_+$, that corresponds to the VBP payment, such that

$$\pi_i^r < B < \pi_i \quad (12)$$

The condition (12) is more likely to hold for small vessels due to $e_i^{r,s} < e_i^{r,l}$ such that $\pi_i^{r,s} < \pi_i^{r,l}$. $\square$

B.5 Proof of Proposition 5

Given that $S_i^r = d_i S_i$, with $d_i = (1 - \lambda_{t_i})$, the necessary condition for $\eta_i^r = \frac{e_{a_i}^r}{S_i^r} < \eta_i = \frac{e_{a_i}}{S_i}$ to hold is $e_{a_i}^r < d e_{a_i}$. Let assume $c_{a_i} - S_i^r < c_{na_i}$, then the benefit function can be any concave function in authorized effort, such that $b_i(e_{a_i} + e_{na_i}) = \alpha e_{a_i}^\beta$. The first-order conditions in (4) can be rewritten before, e_{a_i} , and after, $e_{a_i}^r$, the reform as follows. Before the reform, optimal authorized effort is

$$\frac{\partial b_i(e_{a_i} + e_{na_i})}{\partial e_{a_i}} = \frac{\alpha \beta}{e_{a_i}^{1-\beta}} = c_{a_i} - S_i \quad (13)$$

Expression (13) in terms of optimal authorized effort gives

$$e_{a_i} = \left(\frac{\alpha\beta}{c_{a_i} - S_i} \right)^{\frac{1}{1-\beta}} \quad (14)$$

After the reform optimal authorized effort is

$$\frac{\partial b_i(e_{a_i} + e_{na_i})}{\partial e_{a_i}} = \frac{\alpha\beta}{e_{a_i}^{1-\beta}} = c_{a_i} - d_i S_i \quad (15)$$

Expression (15) in terms of optimal authorized effort gives

$$e_{a_i}^r = \left(\frac{\alpha\beta}{c_{a_i} - d_i S_i} \right)^{\frac{1}{1-\beta}} \quad (16)$$

Given that $e_{a_i}^r < d_i e_{a_i}$, then

$$\left(\frac{\alpha\beta}{c_{a_i} - d_i S_i} \right)^{\frac{1}{1-\beta}} < d_i \left(\frac{\alpha\beta}{c_{a_i} - S_i} \right)^{\frac{1}{1-\beta}} \quad (17)$$

Rearranging the expression (17) to isolate costs and subsidies yields

$$c_{a_i} \left(\frac{1 - d_i^{1-\beta}}{1 - d_i^{2-\beta}} \right) < S_i \quad (18)$$

The expression (18) gives the necessary condition for $\eta_i^r = \frac{e_{a_i}^r}{S_i^r} < \eta_i = \frac{e_{a_i}}{S_i}$ to hold. In other words, pre-reform subsidies must be sufficiently large relative to costs to lower the authorized effort-to-subsidy ratio after the reform, assuming no management measures beyond the VBP.

□